Random Forest Regression

Importing the libraries

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Importing the dataset

In [2]:
```
dataset = pd.read_csv(r"C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and
dataset
```

Out[2]:

	Position	Level	Salary
0	Jr Software Engineer	1	45000
1	Sr Software Engineer	2	50000
2	Team Lead	3	60000
3	Manager	4	80000
4	Sr manager	5	110000
5	Region Manager	6	150000
6	AVP	7	200000
7	VP	8	300000
8	CTO	9	500000
9	CEO	10	1000000

In [3]:
```
X = dataset.iloc[:, 1:2].values
y = dataset.iloc[:, 2].values
```

Splitting the dataset into the Training set and Test set

In [8]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

Feature Scaling

In [10]:
```python
from sklearn.preprocessing import StandardScaler
sc_X = StandardScaler()
X_train = sc_X.fit_transform(X_train)
X_test = sc_X.transform(X_test)
sc_y = StandardScaler()
y_train = sc_y.fit_transform(y_train.reshape(-1, 1))
```

Fitting Random Forest Regression to the dataset

In [11]:
```python
from sklearn.ensemble import RandomForestRegressor
regressor = RandomForestRegressor(n_estimators=15)
regressor.fit(X, y)
```

Out[11]:

RandomForestRegressor

▼ Parameters

	n_estimators	15
	criterion	'squared_error'
	max_depth	None
	min_samples_split	2
	min_samples_leaf	1
	min_weight_fraction_leaf	0.0
	max_features	1.0
	max_leaf_nodes	None
	min_impurity_decrease	0.0
	bootstrap	True
	oob_score	False
	n_jobs	None
	random_state	None
	verbose	0
	warm_start	False
	ccp_alpha	0.0
	max_samples	None
	monotonic_cst	None

Predicting a new result

In [12]:
```
y_pred = regressor.predict([[6.5]])
```

Visualising the Random Forest Regression results (higher resolution)one

In [13]:
```
X_grid = np.arange(min(X), max(X), 0.01)
X_grid = X_grid.reshape((len(X_grid), 1))
plt.scatter(X, y, color = 'red')
plt.plot(X_grid, regressor.predict(X_grid), color = 'blue')
plt.title('Truth or Bluff (Random Forest Regression)')
plt.xlabel('Position level')
plt.ylabel('Salary')
plt.show()
```

```
C:\Users\sande\AppData\Local\Temp\ipykernel_16852\4115633002.py:1: DeprecationWarnin
g: Conversion of an array with ndim > 0 to a scalar is deprecated, and will error in
future. Ensure you extract a single element from your array before performing this o
peration. (Deprecated NumPy 1.25.)
  X_grid = np.arange(min(X), max(X), 0.01)
```